# COMPREHENSIVE CYBERSECURITY LAB PROJECT

## Network Reconnaissance, Web Application Security Analysis & SOC Monitoring


Prepared By:

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Project Domain:

Cybersecurity

Project Type:

Practical Hands-on Lab Project

Lab Environment:

Oracle VirtualBox

Operating Systems:

Kali Linux & Metasploitable 2

Tools Used:

Nmap | WhatWeb | DVWA | Wireshark | Apache Logs

Year: 2026

# ABSTRACT

This project focuses on building and analyzing an isolated cybersecurity lab environment using Oracle VirtualBox, Kali Linux, and Metasploitable 2.

The main purpose of this project is to gain practical knowledge of network reconnaissance, web application security, network traffic analysis, and basic Security Operations Center (SOC) monitoring.

Various cybersecurity tools, including Nmap, WhatWeb, DVWA, and Wireshark, were used to perform security assessments and analyze network communication.

The project also involved examining Apache access and error logs to understand HTTP requests, failed resource requests, and possible reconnaissance-related activity.

Through this hands-on project, I developed a better understanding of how cybersecurity professionals monitor network activity, investigate security events, and identify potential risks.

Project Scope: This work was conducted in an isolated virtual lab for educational and defensive security learning purposes.

# PROJECT OBJECTIVES

The primary objective of this project is to understand practical cybersecurity concepts through a controlled virtual lab environment.

The key objectives are:

**1. Virtual Lab Setup:** Create an isolated cybersecurity lab using Oracle VirtualBox.

**2. Network Reconnaissance:** Identify active hosts and open services using Nmap.

**3. Web Server Fingerprinting:** Identify web server technologies using WhatWeb.

**4. Web Application Security:** Understand SQL Injection concepts and compare DVWA security levels.

**5. Network Traffic Analysis:** Capture and examine HTTP and TCP traffic using Wireshark.

**6. SOC Monitoring:** Analyze network activity and identify relevant security events.

**7. Log Analysis:** Examine Apache access and error logs to investigate HTTP requests and server errors.

**8. Security Recommendations:** Understand defensive measures that can help improve web application and server security.

**Expected Learning Outcome:** Develop practical knowledge of network monitoring, reconnaissance, web application security analysis, and basic SOC investigation techniques.

# LAB ENVIRONMENT AND NETWORK ARCHITECTURE

## 1. Lab Environment

The cybersecurity lab was created using Oracle VirtualBox. Two virtual machines were used to perform practical security testing and monitoring activities.

### Virtualization Platform: Oracle VirtualBox

### Virtual Machines:

- Kali Linux – Used for network reconnaissance, security testing, and traffic analysis.
- Metasploitable 2 – Used as an intentionally vulnerable target machine for educational security testing.

## 2. Network Configuration

Both virtual machines were connected through a VirtualBox Internal Network named "CyberLab".

| Machine | IP Address | Role |
|---|---|---|
| Kali Linux | 192.168.77.20 | Testing Machine |
| Metasploitable 2 | 192.168.77.10 | Target Server |

Network Type: Internal Network

Network Name: CyberLab

## 3. Connectivity Verification

Network connectivity was verified by sending ICMP ping requests from Kali Linux to Metasploitable 2.

The test was successful, confirming communication between both virtual machines within the isolated lab network.

# TOOLS AND TECHNOLOGIES

The following tools and technologies were used during the practical implementation of this cybersecurity project.

## 1. Oracle VirtualBox

Used to create and manage isolated virtual machines for the cybersecurity lab.

## 2. Kali Linux

A Linux-based penetration testing and security assessment operating system used for reconnaissance and network analysis.

## 3. Metasploitable 2

An intentionally vulnerable Linux virtual machine used as the target environment for authorized security testing.

## 4. Nmap

A network scanning tool used to identify active hosts, open ports, and running services.

## 5. WhatWeb

A web technology identification tool used to examine the technologies and server information of a website.

## 6. DVWA (Damn Vulnerable Web Application)

An intentionally vulnerable web application used to understand SQL Injection concepts and compare security configurations.

## 7. Wireshark

A network protocol analyzer used to capture and inspect HTTP and TCP packets.

## 8. Apache Access and Error Logs

Server log files examined to understand incoming web requests, HTTP status codes, and file-not-found events.

Note: All tools were used within the isolated virtual lab for educational and defensive security analysis.

# NETWORK RECONNAISSANCE USING NMAP

Network reconnaissance is an important step in cybersecurity assessment. It helps identify active hosts, open ports, and services running on a target system.

In this project, Nmap (Network Mapper) was used to scan the Metasploitable 2 machine and identify its open ports and running services. The scan was performed within the isolated CyberLab virtual network for educational purposes.

## Nmap Scan Command

The following command was executed in the Kali Linux terminal:

nmap -sV 192.168.77.10

The -sV option is used to detect service versions running on the open ports of the target machine.

## Nmap Scan Results

The Nmap scan successfully detected the target machine at IP address 192.168.77.10.

The scan identified 23 open TCP ports and detected several running services. Some of the important findings were:

- Port 21 – FTP (vsftpd 2.3.4)
- Port 22 – SSH (OpenSSH 4.7p1)
- Port 23 – Telnet

- Port 80 – HTTP (Apache 2.2.8)

- Port 139 and 445 – Samba

- Port 3306 – MySQL 5.0.51a

- Port 5432 – PostgreSQL 8.3

- Port 8180 – Apache Tomcat

The scan also identified other services such as SMTP, DNS, VNC, and UnrealIRCd.

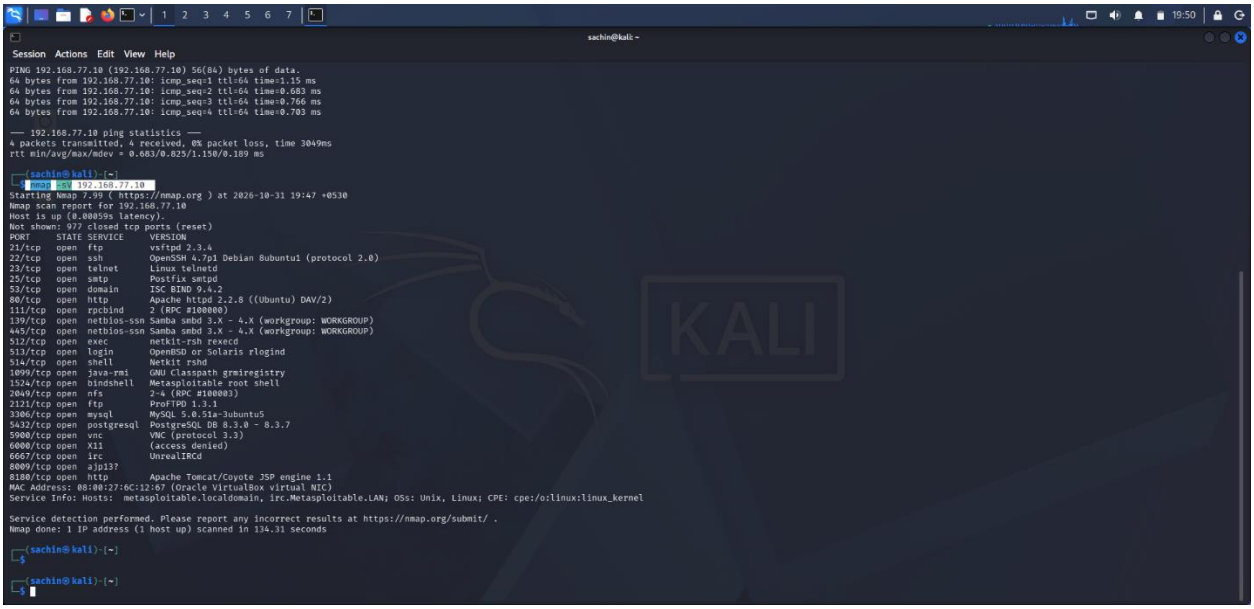



*Figure 1: Nmap Service Version Scan Results of Metasploitable 2*

## Observation

The scan revealed multiple open ports and services on the Metasploitable 2 machine. These findings helped identify the services exposed by the target system. Older service versions may contain known security weaknesses and should be reviewed during a security assessment.

The Nmap scan provided useful information for understanding the target's network exposure and planning further security analysis.

# WEB SERVER FINGERPRINTING USING WHATWEB

WhatWeb is a web technology identification tool used to gather information about websites and web servers. It helps identify technologies such as web servers, programming languages, and other components used by a website.

In this project, WhatWeb was used to examine the web server running on the Metasploitable 2 machine and identify the technologies associated with its web application.

## WhatWeb Command

The following command was executed in the Kali Linux terminal:

whatweb http://192.168.77.10

This command was used to identify the web server and technologies running on the target machine.

## WhatWeb Scan Results

The WhatWeb scan successfully identified the technologies used by the web server running on the target machine.

The scan reported the following information:

- Web Server: Apache 2.2.8
- Operating System: Ubuntu
- Programming Language: PHP 5.2.4
- Web Technology: WebDAV
- HTTP Status: 200 OK
- Website Title: Metasploitable2 - Linux

These results provided information about the web server configuration and the technologies used by the target application.

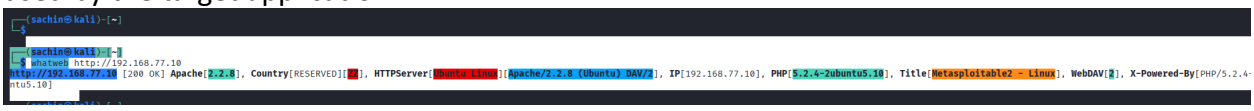



*Figure 2: WhatWeb Technology Identification Results*

## Observation

WhatWeb identified Apache 2.2.8, Ubuntu Linux, PHP 5.2.4 and WebDAV on the target server. The scan helped identify outdated web technologies that may require security updates.

# DVWA – SQL INJECTION SECURITY ANALYSIS

DVWA (Damn Vulnerable Web Application) is an intentionally vulnerable web application used for cybersecurity learning and security testing.

In this project, SQL Injection was studied by comparing DVWA Low and High security levels and examining the application source code.

## Security Level Comparison

**- Low Security:** The application directly uses the user-supplied ID in its SQL query without proper input sanitization.

**- High Security:** The application applies input sanitization and numeric validation before executing the SQL query.

Normal user IDs (1–5) were tested. The application returned the expected records. The source code was examined to understand the security differences. No exploit payloads or data extraction were performed.

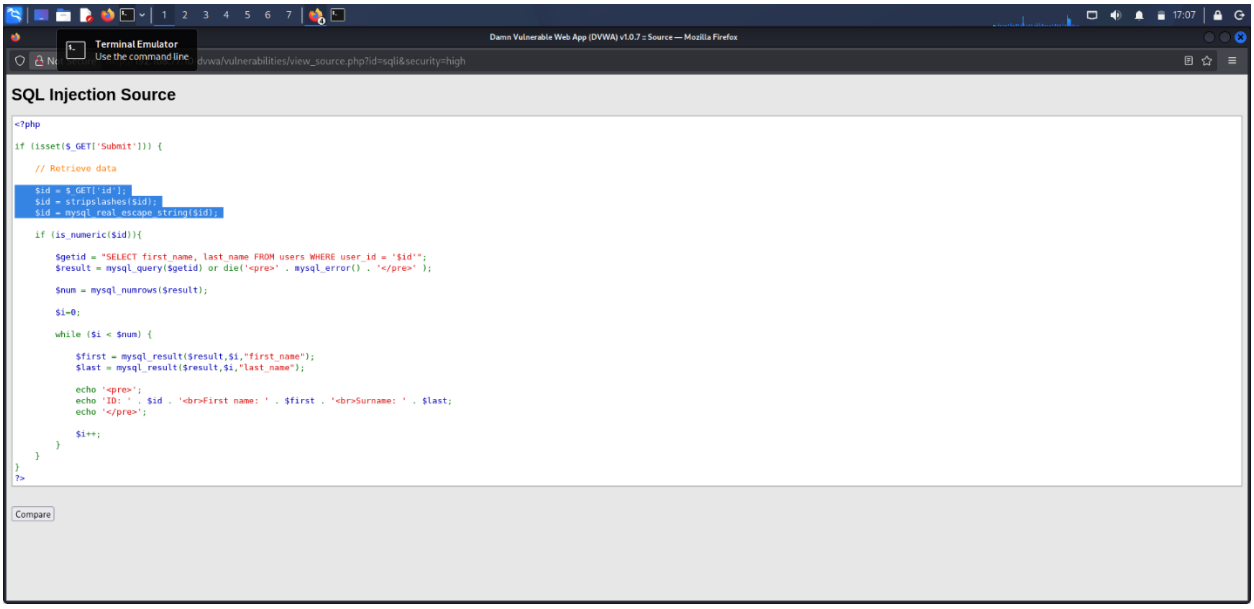



*Figure 3: DVWA High Security Source Code*

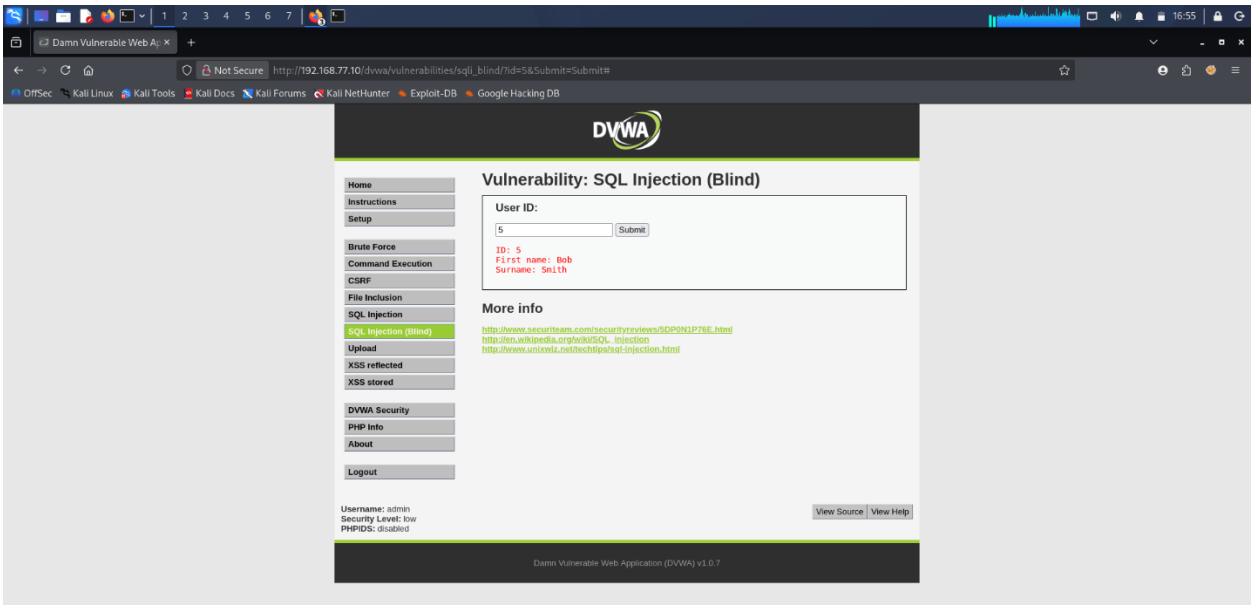



*Figure 4: SQL Injection Normal ID Test Result*

# WIRESHARK - NETWORK TRAFFIC ANALYSIS

Wireshark is a network protocol analyzer used to capture and examine network traffic. In this project, it was used to analyze HTTP requests, server responses, and TCP communication between Kali Linux and  Metasploitable2.

## Wireshark Analysis Results

The following network activities were observed during the analysis:

- HTTP GET requests were captured from Kali Linux to the target server.
- HTTP 200 OK responses confirmed successful requests.
- An HTTP 404 Not Found response was observed during a controlled test.
- TCP SYN, SYN-ACK, ACK, and FIN packets were examined to understand connection establishment and termination.

These observations helped understand normal network communication and basic traffic monitoring.

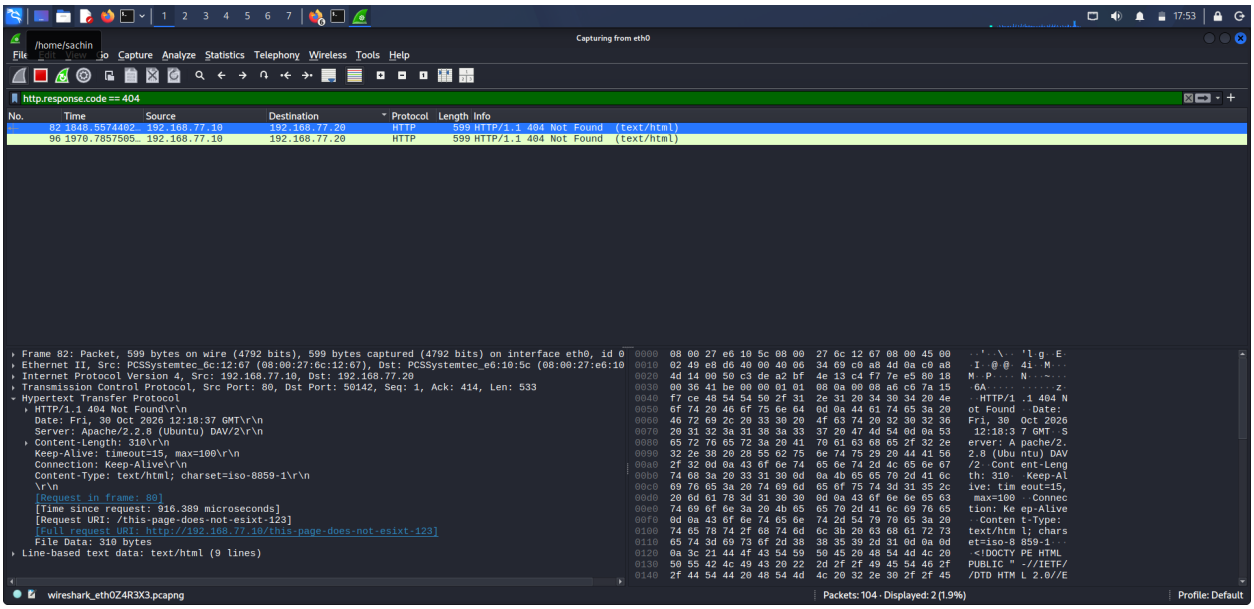



*Figure 5: HTTP 404 Error Response  Analysis*

# SOC MONITORING AND LOG ANALYSIS

Security Operations Center (SOC) monitoring involves observing system and network activities to identify unusual events and potential security issues.

In this project, Apache access and error logs were examined to understand web requests, HTTP errors, and reconnaissance-related activity.

## Apache Log Analysis

Apache access and error logs were examined on the Metasploitable 2 server.

**Key observations:**

- Access logs recorded HTTP requests from Kali Linux ("192.168.77.20").

- Successful DVWA requests returned HTTP 200 status codes.

- Requests to a nonexistent page generated HTTP 404 errors.

- Error logs contained an Nmap-related file-not-found entry.

These logs helped understand how SOC analysts use server logs to investigate web activity and identify reconnaissance indicators.

```
192.168.77.20 - - [30/Oct/2026:08:20:39 -0400] "GET /this-page-does-not-esixt-12
3 HTTP/1.1" 404 310 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/2010010
1 Firefox/140.0"
msfadmin@metasploitable:~$ sudo tail -n 20 /var/log/apache2/error.log
[Sun May 20 15:56:39 2012] [notice] caught SIGWINCH, shutting down gracefully
[Fri Oct 02 02:50:20 2026] [notice] Apache/2.2.8 (Ubuntu) DAV/2 configured -- re
suming normal operations
[Fri Oct 02 03:03:15 2026] [notice] caught SIGWINCH, shutting down gracefully
[Fri Oct 30 06:39:26 2026] [notice] Apache/2.2.8 (Ubuntu) DAV/2 configured -- re
suming normal operations
[Fri Oct 30 06:51:52 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/nmaplowercheck1793357512
[Fri Oct 30 06:51:52 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/sdk
[Fri Oct 30 06:51:52 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/evox
[Fri Oct 30 06:51:52 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/HNAP1
[Fri Oct 30 07:03:13 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/favicon.ico, referer: http://192.168.77.10/
[Fri Oct 30 08:18:37 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/this-page-does-not-esixt-123
[Fri Oct 30 08:20:39 2026] [error] [client 192.168.77.20] File does not exist: /
var/www/this-page-does-not-esixt-123
msfadmin@metasploitable:~$ _
```

*Figure 6: Apache Error Log Analysis*

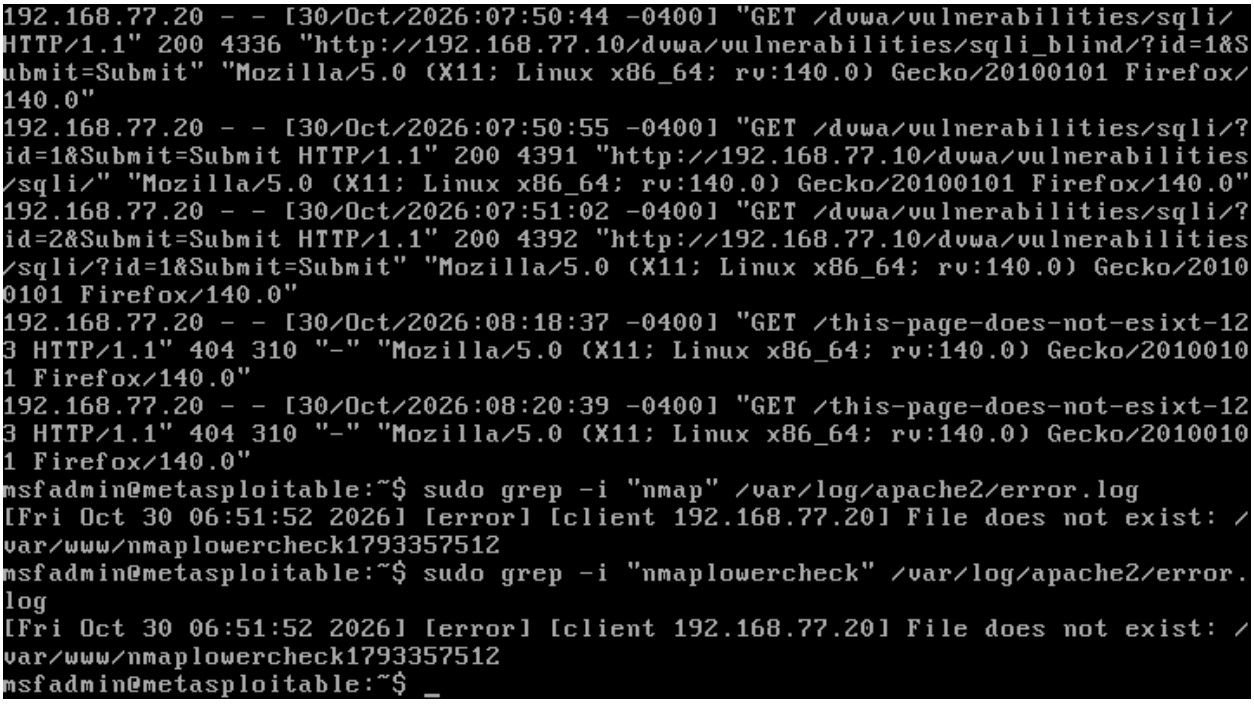



*Figure 7: Nmap-Related Activity Detected in Apache Logs*

## SECURITY RECOMMENDATIONS

Based on the lab observations, the following security measures are recommended:

1. Keep web servers and software updated.

2. Use secure coding practices and prepared SQL statements.

3. Monitor server logs for unusual requests and errors.

4. Use HTTPS to protect sensitive web traffic.

5. Restrict unnecessary open ports and services.

## CONCLUSION

This project provided practical experience in network reconnaissance, web application security analysis, network traffic monitoring, and basic SOC investigation.

By using Nmap, WhatWeb, DVWA, Wireshark, and Apache logs, I learned how to identify exposed services, examine web technologies, analyze network communication, and investigate server events. This hands-on project improved my understanding of cybersecurity tools and basic security monitoring.